Safety and Security

Definition

SECURITY vs. SAFETY (Leveson, 1986)

- **Security** : protect the **system** from the **environment**; **threats**, **intentional**. Methods : vulnerability scans, ethical hacking, audits, code review.
- **Safety** : protect the **environment** from the **system**; **hazards**, **accidental**. Methods : testing, failure mode analysis, redundancy.

Both are about **risks** (probability of something bad).

Example

- Car misreads 10% of signs : **safety hazard**.
- Stickers make a stop sign read as a speed limit : **security threat** + **safety hazard**.

Socio-technical system : social and technical systems affect each other (social media, autonomous cars) : hidden loops, emergence, non-determinism.

Explained to a goat

Security : stop the wolf breaking the fence (on purpose). Safety : stop the fence falling on the goats (by accident).

Four Scenarios of Harm

Definition

WHEN SOFTWARE CAUSES HARM (Parasuraman & Riley)

- **Malfunction** : does not work as specified, unintentional bug (Therac-25, CrowdStrike 2024).
- **Abuse** : malicious use by bad actors (ransomware on hospitals).
- **Misuse** : unintended / unsupported use, careless (Samsung code in ChatGPT).
- **Intended use** : used as planned and still harms (infinite scroll, elections); the hardest to anticipate.

Warning

Misuse vs. abuse : the difference is **intent**. Viral travel trends ruining ecosystems : **intended use**.

Explained to a goat

It breaks, a villain uses it, someone uses it wrong, or it works perfectly and still hurts.

User-Generated Content

Definition

UGC : content (text, images, video) posted publicly by users on platforms.
Harms (Banko 2020) : hate & harassment, self-inflicted harm, ideological harm, exploitation. **Toxic content** : harmful or offensive material.

Definition

HATE SPEECH (UN 2019) : discriminatory or pejorative + about identity factors (religion, ethnicity, gender...) + any form of expression.

Example

Facebook and the Rohingya (Myanmar 2017) : hate and military disinformation, weak and slow moderation.

Key idea

Free speech (UDHR art. 19) is not absolute. Filtering errors :

- **FN** : toxic content escapes (people harmed);
- **FP** : harmless content censored (free speech harmed).

Who decides? States, platforms, people.

Explained to a goat

Delete too little and people get hurt; delete too much and innocent voices are silenced.

Content Moderation

Definition

CONTENT MODERATION (Grimmelmann 2015) : governance mechanisms structuring participation to facilitate cooperation and prevent abuse = **identification** + **action**.

Key formulas

When : pre-publication; post-publication (systematic, sampling, **reactive** on reports).
How :

- **Classification** : keyword screening, sentiment analysis, ML classifiers, computer vision.
- **Matching** : hashing vs. database; **perceptual hashing** robust to small changes (Christchurch video : 80% of re-uploads blocked).

Actions : **exclude** (ban, timeout) > **organize** (delete, annotate, reduce visibility) > **set norms** (warn).

Warning

Automation misses context and nuance; users evade filters ⇒ **hybrid**. Human moderators face severe mental health risks : a **dilemma** (some exposed to protect others).

Explained to a goat

Find bad posts (by type or by matching known ones), then ban, delete, hide or warn. Machines miss jokes; the humans who help get hurt.

The Confusion Matrix

Key formulas

Rows = **actual**, columns = **predicted** :

	pred. +	pred. −
actual +	TP	FN
actual −	FP	TN

$$\text{Acc} = \frac{TP + TN}{\text{all}}$$

$$\text{FNR} = \frac{FN}{TP + FN} \qquad \text{FPR} = \frac{FP}{TN + FP}$$

Example

Fissures (positive = fissure) : TP 60, FN 15, FP 20, TN 100. FNR $= 15/75 = 20\%$. For safety, watch **FN** (missed fissures).

Warning

Check what **positive** means. COVID test : FN dangerous. Quality control with positive = "good" : FP dangerous.

Explained to a goat

False alarm (FP) or missed wolf (FN) : which is worse depends on the story.

Strategy : Bad Actors

Definition

BAD ACTOR : person or group with **malicious intent** exploiting a system ; often external, also insiders.

Key formulas

5 motivations (they overlap) :

- **Money** : scammers, hackers, competitors (ransomware, data theft).
- **Politics** : nation states, agencies (disinformation, infrastructure attacks).
- **Entertainment** : trolls, thrill seekers.
- **Ideas** : hacktivists, extremists, conspiracy theorists (DoS, leaks, defacement).
- **Self-interest** : revenge, obsession, insiders (harassment, tracking, grade changing).

Warning

Graded : **intent** explicit + **motivation** explicit + chain **action** → **impact on stakeholders**, specific and diverse.

Explained to a goat

Imagine the villains, why they come, what they do, who gets hurt.

Strategy : STRIDE

Key formulas

- **S**poofing (pretend to be another) → authenticity
- **T**ampering (covert modification) → integrity
- **R**epudiation (deny, no proof) → accountability
- **I**nformation disclosure → confidentiality
- **D**enial of service (overload) → availability
- **E**levation of privilege → authorization

Explained to a goat

Six burglar tricks : disguise, secret changes, "wasn't me", peeking, blocking the door, stealing the master key.

Strategy : Harm Modeling

Key formulas

1. Category of use (malfunction, abuse, misuse, intended use). **2. Type of harm**. **3. Magnitude** : severity, scale, probability, frequency. Iterative, every stage.

Definition

TAXONOMY (what is harmed)

- **Humans** : physical injury ; emotional / psychological injury.
- **Allocation of resources** : opportunity loss ; economic loss.
- **Human rights** : dignity, liberty, privacy loss ; environmental impact.
- **Social systems** : manipulation ; social detriment.

Example

Post unfairly censored : **liberty**. GPS leak : **privacy**. Recruitment discriminates on names : **opportunity**. "Nurse" shows only women : **social detriment**.

Explained to a goat

Go through "bodies, minds, money and chances, rights, society" and ask : can my app hurt this, how badly, how widely, how often ?

Weekly Summary

Summary

- Security (threats, intentional) vs. safety (hazards, accidental).
- Malfunction, abuse, misuse, intended use.
- UGC harms, hate speech ; FN vs. FP dilemma with free speech.
- Moderation : when, classification vs. matching, exclude / organize / set norms.
- Confusion matrix, accuracy, FNR, FPR.
- Bad actors (5 motivations), STRIDE, harm modeling (10 types).